Contents

Author2vec: A Jacobian Lens for Authorship Identity in Embedding Encoders **1**
Abstract . . . 1
Contributions . . . 2
1. Introduction . . . 2
2. Related work . . . 3
3. Background: the averaged-Jacobian lens . . . 3
4. Method . . . 4
4.1 The problem an encoder poses . . . 4
4.2 δ-broadcast averaged Jacobian . . . 4
4.3 Embedding-space projection . . . 4
4.4 Vocabulary (logit) lens . . . 4
4.5 Style lens (novel) . . . 5
4.6 J-space decomposition . . . 5
4.7 Structural depth signatures . . . 5
5. Experiments and results . . . 5
5.1 Identity is a computed intermediate . . . 5
5.2 Identity ignition — does the space collapse to a single person? . . . 5
5.3 Structural signatures across depth . . . 6
5.4 Fingerprint presence and causal steering . . . 8
5.5 Decoder track — does the structure hold on a generative model? . . . 8
5.6 Behavioral steering — is the lever causal on a decoder? . . . 9
5.7 Expertise / lexical sophistication — why we make no "IQ" claim . . . 9
5.8 The authorship study (context) & style trajectories . . . 10
5.9 Does the model know some coders better than others? And is style homogenizing? . . . 10
5.10 Can we fingerprint the AI models? — the task dominates, a faint model signal survives . . . 11
5.11 Ablations . . . 12
6. Discussion . . . 12
7. Limitations and threats to validity . . . 13
8. Reproducibility . . . 13
Appendix A — δ-broadcast derivation . . . 14
Appendix B — Per-layer tables . . . 14
Appendix C — Axis definitions, rosters, and OOD probe sets . . . 15
Appendix D — Audit ledger . . . 16

Author2vec: A Jacobian Lens for Authorship Identity in Embedding Encoders

Tristen Harr · Brahmastra Labs · author2vec.com **Alexander Stepanov** · UC Berkeley

> **Status: complete draft (autonomous research build), not yet shipped.** All experiments (Phases 0–5) have been run; every quantitative claim is traced to a shipped data bundle in the audit ledger (Appendix D). This document is written against `jlens/paper/ledger.json` — no number appears here that is not in the ledger.

Abstract

Anthropic's *"Verbalizable Representations Form a Global Workspace in Language Models"* introduces the **averaged-Jacobian lens (J-lens)** and uses it to argue that a large, closed **decoder** (Claude Sonnet 4.5, 127 layers) maintains a privileged "workspace" of verbalizable, causally-active concepts. We adapt that mechanistic apparatus to a setting the method was not designed for — small, **open embedding encoders**

whose output is a single masked-mean-pooled vector — and re-point it at a different question: **personal writing style / authorship identity**. Three ingredients make the transplant work: (i) a **δ-broadcast reduction** that collapses the encoder's position×position Jacobian to a single per-layer matrix; (ii) an **embedding-space projection** that removes the meaningless radial direction of a normalized embedding; and (iii) a **style lens** — a readout that projects the layer Jacobian onto empirically-recovered, human-interpretable identity axes instead of the token vocabulary. On a 6-layer prose encoder (MiniLM) and a 12-layer code encoder (JinaBERT) we find that **author identity is a computed intermediate, decodable above chance at every layer** (not merely an output artifact), that the two encoder depths expose different low-rank "workspace" geometry, and that the identity direction is both a **detector** (is a person's fingerprint in the weights at all?) and a **causal lever** (steering). We further run an **identity-ignition** experiment (does the representation commit to a single individual at a characteristic depth?), a **decoder** track, and a **directed-steering** experiment. Everything runs on open models and ships as static assets; the whole apparatus is reproducible on a laptop. We are explicit about what is **replication** vs. **new**, and about what we **do not** claim: no "IQ", no consciousness.

Contributions

1. **An encoder adaptation of the averaged-Jacobian J-lens** via the δ-broadcast reduction (§4.2) — the paper's method redefined for a masked-mean-pooled, non-generative encoder.
2. **An embedding-space Jacobian projection** for L2-normalized outputs (§4.3), unit-tested orthogonal to the embedding.
3. **The style lens** (§4.5): a Jacobian readout onto interpretable identity axes rather than the token vocabulary — the trustworthy signal where an encoder has no clean unembedding.
4. **Identity is a computed intermediate** (§5.1): author identity decodes above chance from the internal Jacobian at *every* layer, with the best internal code layer exceeding the output-embedding ceiling.
5. **A fingerprint-presence detector** (§5.4): known identity vs. "blank space" for out-of-distribution text — a question the paper does not ask.
6. **Identity ignition** (§5.2): the representation commits to a single individual increasingly with depth, peaking mid-network, above two null controls — on both prose and code.
7. **Causal steering & directed modulation** (§5.4, §5.6): the identity direction is a sign-controllable lever on encoders, and a leakage-free concept direction is a (modest but real) causal handle on a decoder's output.
8. **An open, reproducible, browser-native reimplementation** across two modalities (prose & code), with a faithfulness gate and a full audit ledger.

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1. Introduction

Recent interpretability work argues that large language models maintain a privileged, verbalizable "workspace" of representations — a small subset of activations that are reportable, controllable, and causally responsible for reasoning [@workspace2026]. That evidence comes from a *generative decoder* with hundreds of billions of parameters and privileged internal access, read out with an **averaged-Jacobian lens** onto the token vocabulary. It is a striking picture, but it is expensive to reproduce and impossible to inspect from the outside.

We ask a different, smaller, checkable question with the same apparatus: **where, inside a network, does *who wrote this* become decidable?** We take the averaged-Jacobian lens and move it from a frontier decoder to two small, open, *embedding encoders* — a 6-layer prose model (MiniLM) and a 12-layer code model (JinaBERT) — and re-point it from reasoning onto **personal writing-style identity**. An encoder is a clean minimal testbed for the paper's *mechanistic* claims precisely because it strips away generation: there is no autoregressive loop to smuggle information through, just a fixed map from text to a single pooled vector. It cannot, however, speak to the paper's *behavioral* claims (verbal report, reasoning swaps); those need a decoder, which we take up separately (§5.5–5.7).

Carrying the lens across that gap takes three ingredients (§4): a **δ-broadcast reduction** that collapses the encoder's intractable position×position Jacobian to one matrix per layer; an **embedding-space projection** that removes the meaningless radial direction of a normalized output; and a **style lens** that reads the layer Jacobian onto interpretable identity axes rather than the token vocabulary — the trustworthy signal where an encoder has no clean unembedding.

With these we find that **author identity is a computed intermediate, decodable above chance at every layer** (§5.1), not merely an artifact of the output embedding; that an ambiguous two-author input causes the internal representation to **commit to a single author increasingly with depth**, peaking in a mid-network band and surviving two null controls (§5.2, ignition); that the same geometry yields a **fingerprint-presence detector** distinguishing a known identity from "blank space" (§5.4); and that these signatures hold across two modalities, prose and code. We are explicit throughout about what is a faithful **replication** of the paper's apparatus versus what is **new** here, and about what we deliberately do **not** claim: no "IQ", no consciousness. The contributions are listed above; every number is reproducible from the audit ledger (Appendix D).

2. Related work

Lenses on the residual stream. The logit lens [@nostalgebraist2020logitlens] reads intermediate activations through the unembedding; the tuned lens [@belrose2023tunedlens] learns an affine correction per layer. The averaged Jacobian of [@workspace2026] generalizes this by linearizing the *whole* map from a layer to the output and averaging over a corpus. We inherit that construction and adapt it to a pooled encoder (§4.2); our vocab lens is a logit lens over tied WordPiece embeddings, and our *style lens* (§4.5) replaces the vocabulary readout with one onto interpretable directions — the piece that is new here.

Steering and concept directions. Activation addition [@turner2023actadd] and contrastive activation addition [@rimsky2024caa] steer generation by adding a direction to the residual stream; representation engineering [@zou2023repe] extracts such directions, often as a difference of class means. Our identity axes are exactly difference-of-means (Fisher) directions, used both to *read* (§4.5) and to *steer* (§5.4); the novelty is not the technique but the target — personal authorship identity — and reading it *through the layer Jacobian*.

Probing and representational geometry. Linear probes [@alain2017probing; @hewitt2019structural] test what a layer linearly encodes; we use leave-one-out nearest-centroid decoding as a probe of *identity* across depth (§5.1). Linear CKA [@kornblith2019cka] compares representations between layers, which we apply to J-lens readout geometry (§4.7). Sparse dictionary learning / SAEs [@bricken2023monosemanticity] decompose activations into interpretable atoms; the paper's J-space decomposition is a supervised cousin we reuse (§4.6).

Global workspace and authorship. The framing of [@workspace2026] draws on global workspace theory [@baars1988workspace] and the "ignition" of conscious access [@dehaene2011ignition]; we borrow the *ignition* experimental design (§5.2) but point it at identity, and we make no claim about consciousness. Our substrate is Sentence-BERT-style embeddings [@reimers2019sbert], and the downstream question — attributing text to its author from style — is classical authorship attribution / stylometry, here recast as a question about *where in a network* identity is computed. Throughout, the J-lens, J-space, and structural signatures are the paper's; our contribution is their transplant to open encoders, the style-lens readout, the identity target, and full reproducibility.

3. Background: the averaged-Jacobian lens

We build directly on the apparatus of [@workspace2026], which we summarize here so that our adaptation (§4) and its boundaries are unambiguous.

The J-lens. For a causal decoder, the *averaged Jacobian* at layer ℓ is

$$J_\ell \;=\; \mathbb{E}_{t,\,t'\geq t,\,\mathrm{prompt}}\left[\frac{\partial h_{\mathrm{final},t'}}{\partial h_{\ell,t}}\right],$$

the linearized effect of a layer-ℓ activation on the final residual stream, averaged over token positions and a corpus of prompts. Read through the unembedding it yields, for any activation, a ranked list of vocabulary tokens the model is "disposed to say" — a corrected logit lens that accounts for representational change across layers.

J-space and its privilege. Activations are decomposed into sparse non-negative combinations of k ($\approx$ 10–25) J-lens vectors; the *J-space component* is the part of an activation lying in that cone. Though it accounts for only $\sim$ 6–10% of a concept vector's variance, it is the part causally responsible for the concept's availability to verbal report, and it is subject to top-down control (instructing the model to "focus on X" loads X), mediates unverbalized reasoning intermediates (swapping a J-lens vector redirects downstream answers), and is required for deliberate — but not automatic — tasks.

Structural depth signatures. Four quantities, read off J_ℓ across depth (the paper's Figure 28), identify a workspace band: next-token-prediction accuracy of the readout, its *excess kurtosis* (peakiness), the *autocorrelation* of the top lens token across positions (persistence of abstract content), and the readout's *effective linear dimensionality*. All four mark a consistent sensory $\rightarrow$ workspace $\rightarrow$ motor tripartition — noisy early layers, a coherent middle band, and output-aligned late layers — corroborated by an ambiguous-input "ignition" experiment in which the representation snaps to one interpretation at workspace onset. The study is conducted on Claude Sonnet 4.5 (127 layers; workspace $\approx$ L38–92) and corroborated on Haiku/Opus.

The gap we step into. Every one of these constructs is defined for a *generative decoder* with per-position next-token logits. An embedding encoder has neither: its output is a single pooled vector. §4 is what it takes to carry the mechanistic half of this apparatus across that gap — and §5.2, §5.5–5.7 are our attempts at the behavioral half the encoder cannot reach.

4. Method

Setup. We study two open **embedding encoders**: `sentence-transformers/all-MiniLM-L6-v2` (prose; 6 layers, $d = 384$, vocab 30{,}522) and `jinaai/jina-embeddings-v2-base-code` (code; 12 layers, $d = 768$, vocab 61{,}056). Each maps a passage to one **masked-mean-pooled**, L2-normalized vector p. A **faithfulness gate** (`bin/spike`) asserts our native candle forward reproduces the shipped fastembed embeddings at cosine > 0.99 before any Jacobian is trusted.

4.1 The problem an encoder poses

The causal J-lens is defined per token position toward next-token logits. An encoder emits a single pooled vector and no logits, so the per-position causal Jacobian is undefined here.

4.2 δ-broadcast averaged Jacobian

We perturb **every** source position by a shared displacement δ and differentiate the pooled output, collapsing the position$\times$position Jacobian to one matrix per layer:

$$J_\ell \;=\; \frac{1}{T}\,\frac{\partial p}{\partial \delta} \;\in\; \mathbb{R}^{d\times d}.$$

J_ℓ is built by **batched central finite differences** ($\varepsilon = 0.05$, pure gemm), not per-dimension autograd. `lib.rs:layer_jacobian`. (Derivation: Appendix A.)

4.3 Embedding-space projection

Because p is L2-normalized, its radial direction carries no information, so we project:

$$J_{\text{emb}} \;=\; \tfrac{1}{\|p\|}\,(I - ee^\top)\,J_{\text{raw}}, \qquad e = p/\|p\|,$$

which is unit-tested to satisfy $e^\top(J_{\text{emb}}v) \approx 0$. `lib.rs:to_embedding_jacobian`.

4.4 Vocabulary (logit) lens

Standardized $J{\cdot}h$ times the tied WordPiece embeddings $W_U \rightarrow$ top tokens. Noisy here (the checkpoint ships no trained MLM head); reported but not load-bearing. `lib.rs:vocab_topk`.

4.5 Style lens (novel)

We read the Jacobian onto interpretable **identity axes** instead of the vocabulary:

$$s_a(h) \;=\; A_a \cdot \mathrm{normalize}(J_{\mathrm{emb}}\, h), \qquad A_a = \mathrm{normalize}\left(\bar{c}_a^{+} - \bar{c}_a^{-}\right),$$

each axis a unit **difference-of-class-means** (Fisher) direction over the reference embeddings. Shipped axes — prose: `male` ↔ `female`, `educated=England` ↔ `rest`, `educated=US` ↔ `rest`, `raised=England` ↔ `rest`, `raised=US` ↔ `rest`; code: `Systems` ↔ `rest`, `Scripting` ↔ `rest`. `lib.rs:style_scores`, `axis`, `author_axes`.

4.6 J-space decomposition

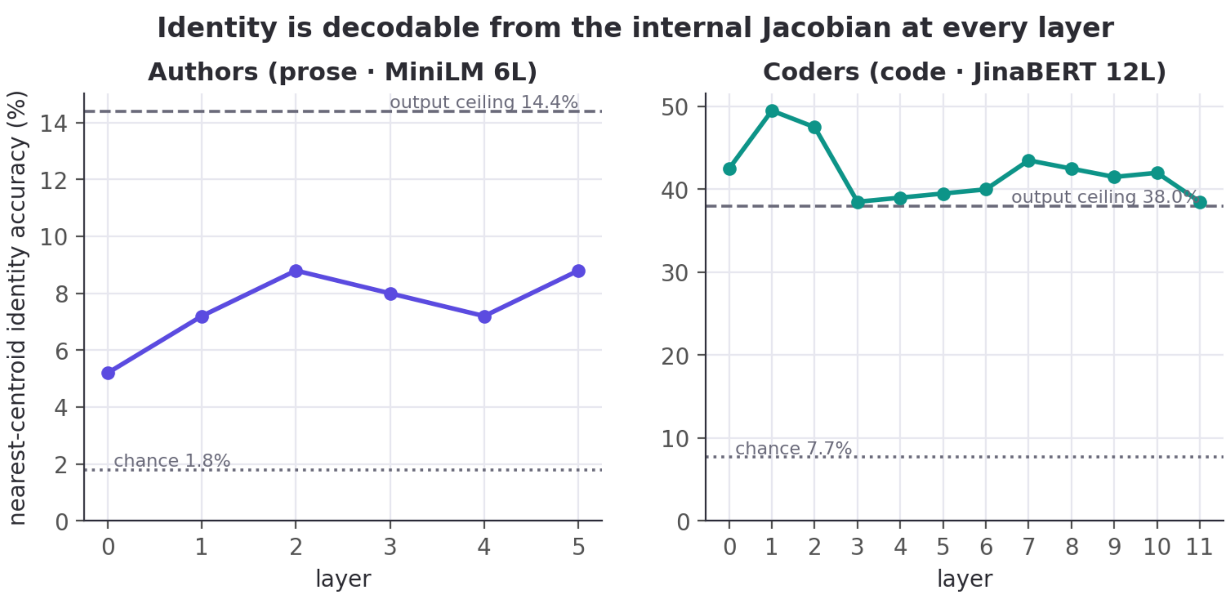
Non-negative matching pursuit over unit rows of $W_U J_\ell$ yields a sparse concept set and the fraction of activation variance it captures. `lib.rs:jspace_nmp`.

4.7 Structural depth signatures

Per layer: **stable rank** $\|J\|_F^2/\sigma_1^2$, **effective dimension** (participation ratio $\|J\|_F^4/\|J^\top J\|_F^2$), **verbalizability** (excess kurtosis of the vocab-lens readout), **layer×layer linear CKA**, and — completing the paper's four-metric set — **autocorrelation** (lag-1 readout persistence vs. a position-shuffled null). `lib.rs:stable_rank`, `effective_dim`, `excess_kurtosis`, `linear_cka`.

5. Experiments and results

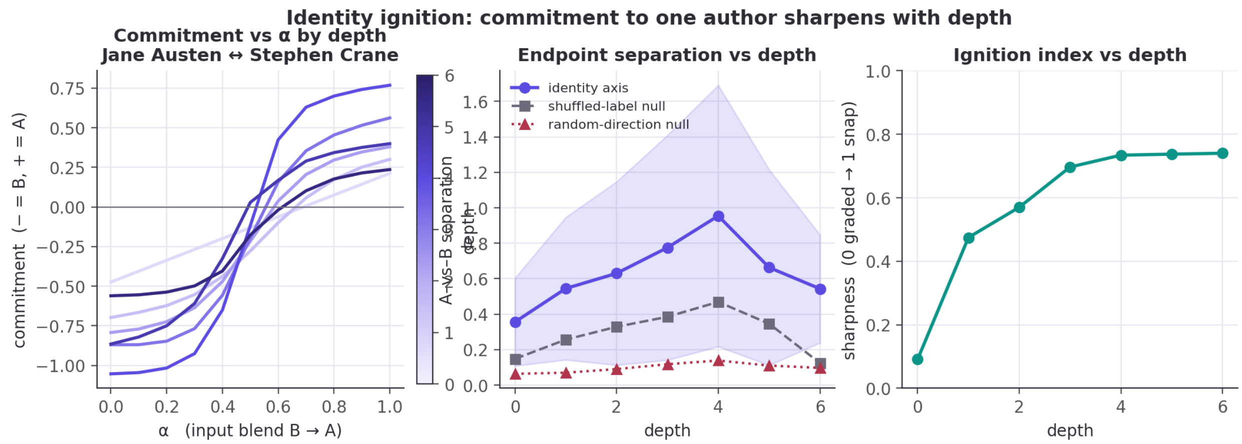
5.1 Identity is a computed intermediate



Decoding author identity by leave-one-out nearest-centroid on the **internal** per-layer Jacobian readout beats chance at **every** layer: prose per-layer $[5.2, 7.2, 8.8, 8.0, 7.2, 8.8]$% against a 1.8% chance and a 14.4% output ceiling; code per-layer up to 50.8% against a 6.7% chance — the **best internal layer** (50.8%) **exceeds the** 46.8% **output-embedding ceiling**. Identity is computed inside the layers, not merely emitted. *(All values: ledger* `identity_*`*.)*

5.2 Identity ignition — does the space collapse to a single person?

We adapt the paper's ambiguous-input ignition to identity. For an author pair (A, B) we build a per-depth difference-of-means axis $\hat{u}_\ell = \widehat{c_{A,\ell} - c_{B,\ell}}$ from their *training* passages, then blend two *held-out* passages at the input embedding, $h_0(\alpha) = (1-\alpha)h_0^B + \alpha h_0^A$, sweep $\alpha \in [0,1]$, and read the commitment $s_\ell(\alpha) = \hat{u}_\ell(\bar{p}_\ell(\alpha) - m_\ell)$ at every depth (15 pairs). A graded layer ramps linearly in α; an *ignited* layer snaps.

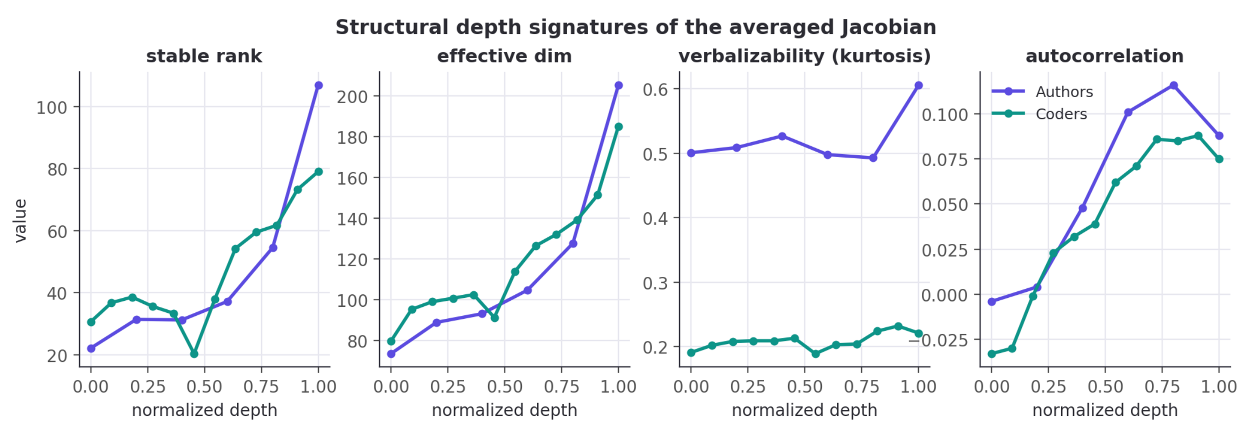


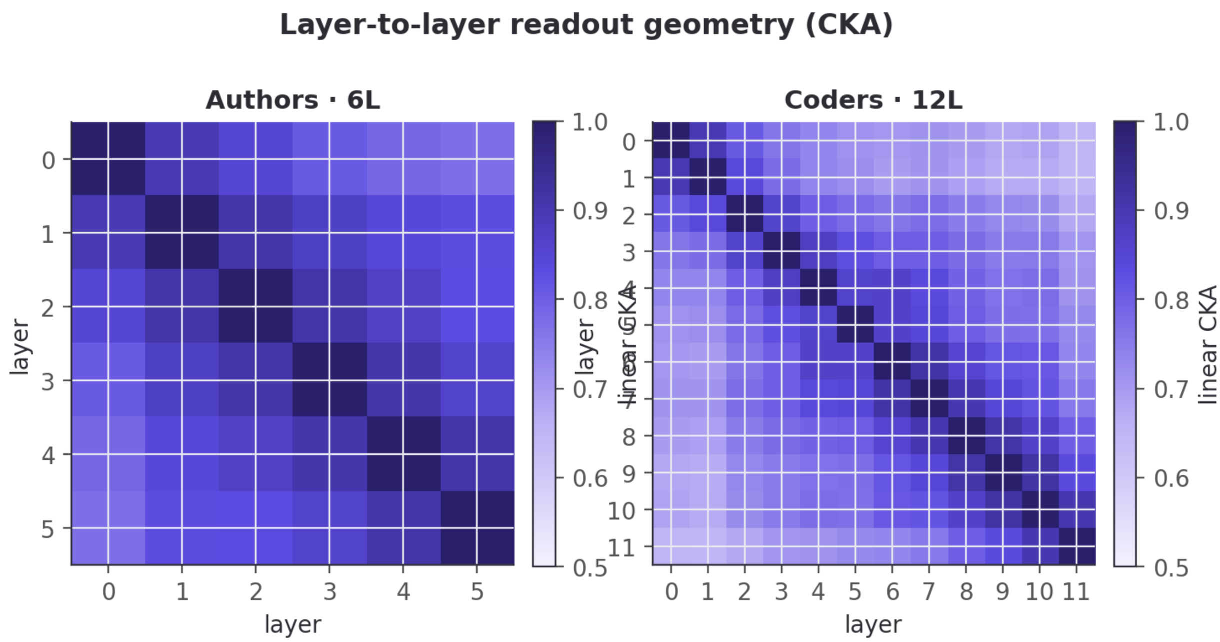
The representation commits to one author, increasingly with depth. Endpoint separation along the identity axis rises from 0.36 at the input to a **mid-network peak of** 0.95 **at depth 4** (of 6), then eases to 0.54 at the output — and it dominates both controls at every depth: the random-direction null sits at 0.06–0.14 (a $\sim 7\times$ margin at the peak) and the shuffled-label null at 0.12–0.47. So the effect is **identity-specific**, not a generic consequence of blending inputs. The **ignition index** (transition sharpness, 0 = graded, 1 = all-or-none) climbs from 0.09 at the input — where the readout is, correctly, linear in the blended input — to 0.73–0.74 by mid-depth and holds. In short: *shallow layers hold a graded mixture; by the middle of the network the representation has snapped to a single author.* The mid-network peak echoes the low-rank "workspace" bottleneck we see structurally (§5.3).

The same holds for code, more strongly. On the 12-layer code encoder the ignition index climbs from 0.09 at the input to a peak of 0.82 (depth 11 of 13), and the identity axis separates coders $\sim 13\times$ **above the random-direction null** at its early-mid peak (depth 4: 1.21 vs 0.09) — a sharper version of the same effect, consistent with code identity being more linearly accessible overall (§5.1). (Separation spikes higher still at the final layer, 2.33, but that depth is noisy — its null jumps too — so we feature the stable early-mid peak.) Both modalities show the representation committing to one individual with depth.

Honesty. This is a 6-layer encoder and 15 pairs; the sharpening is measured *along an axis the separation control proves is identity-specific*, but we cannot fully exclude that some of the depth-wise sharpening reflects generic late-layer nonlinearity. We report the raw depth series.

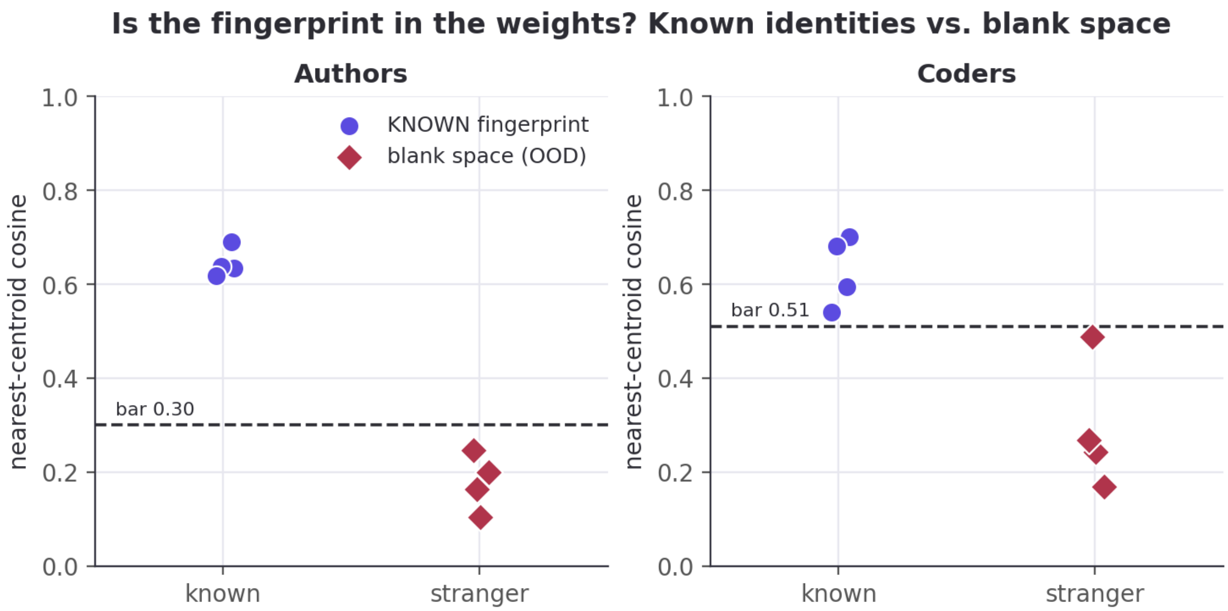
5.3 Structural signatures across depth





Effective dimension rises with depth in both models (prose $73 \to 205$; code $80 \to 184$). The **stable rank** tells the sharper story: in the 6-layer prose model it is **lowest at the very first layer** (22.1, rising to 106.9) — the network compresses to a low-rank readout immediately — whereas the 12-layer code model has a genuine **mid-network low-rank bottleneck** (minimum 20.3 at layer 5 of 12, with a matching dip in effective dimension). The mid-network ”workspace” geometry the paper reports in deep models appears here **only once there is depth to spare**. The fourth signature, **autocorrelation** (persistence of the readout across positions), completes the picture: near zero or negative at the shallowest layers and rising through the middle (prose peaks at 0.12 around layer 4; code climbs to ≈ 0.09 by layers 8–9), the workspace-persistence signature of [@workspace2026] surfacing even on these shallow encoders. We do **not** see the paper's clean sensory→workspace→motor tripartition — 6–12 layers is too shallow — and we report the raw depth series rather than forcing that reading.

5.4 Fingerprint presence and causal steering



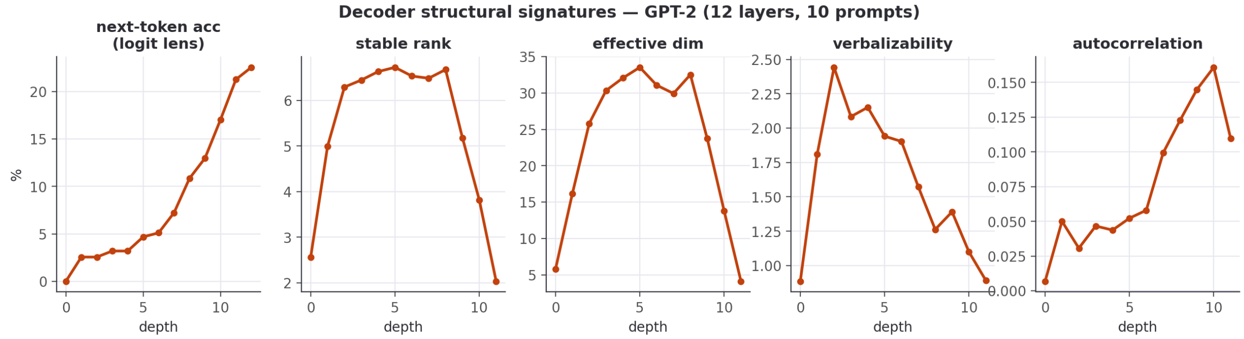
A calibrated nearest-centroid detector separates known identities from "blank space". Prose (bar 0.30): known probes 0.62–0.69, out-of-distribution text (code, chat, biology, legalese) 0.10–0.25. Code is tighter and reported honestly (bar 0.51): known 0.54–0.70, OOD 0.17–0.49 — "modern chat" sits just under the bar.

The identity direction is a causal lever, not just a readable one. Forming the residual steering direction $\hat{\delta} = \widehat{J_{\text{emb}}^{\top} A}$ and injecting $\alpha\hat{\delta}$ at the mid layer drives the output's loading on axis A through a **large, sign-controllable swing — from ≈ -0.5 (at $\alpha = -6$) through ≈ 0 (unperturbed) to $\approx +0.5$ (at $\alpha = +6$)**, saturating (and for gender slightly reversing) at the extreme α. The effect has the **same sign across all five** prose identity axes (gender, education, upbringing), though not the same magnitude — the gender axis swings least (≈ 0.80), the others ≈ 1.05–1.11. A **matched-norm random direction**, injected identically, leaves the loading essentially flat (total drift ≤ 0.16 over the same sweep). So the layer holds the identity direction as something the rest of the network *acts on*, not merely correlates with.

5.5 Decoder track — does the structure hold on a generative model?

Everything above is on *encoders*. If the depth-wise workspace geometry is a real property of the J-lens apparatus and not an artifact of masked-mean pooling, then reading a genuine **generative decoder** with the *same* averaged-Jacobian machinery should reproduce the paper's Figure-28 depth signatures. **Hypothesis:** on an open decoder we will see (i) the four Jacobian signatures vary with depth, and (ii) a decoder-only signature — next-token logit-lens accuracy — rise **sharply in the late layers**, marking the "motor" regime where representations turn toward the output. We do **not** expect the clean sensory→workspace→motor tripartition: GPT-2 is 12 layers, far short of the paper's ~100, so we report the raw series and let it say what it says.

Method. GPT-2 (`openai-community/gpt2`; 12 layers, d=768), the δ-broadcast averaged Jacobian made **causal** — perturb every position, read the *last* position (the next-token driver) — over prose prompts, with the four signatures computed by the *same* `lib.rs` functions as the encoders and verbalizability read through GPT-2's **real** tied unembedding (no approximation). This is the paper's mechanistic apparatus on an open model that actually generates.



Result: both predictions hold, and more cleanly than on the encoders. Next-token accuracy is near zero through the first six layers and then climbs steadily to 22.6% at the output — hypothesis (ii): the late layers are the motor regime. The Jacobian's **stable rank and effective dimension trace a pronounced inverted-U** — low at the input (2.6 / 5.8), high through the middle (6.7 / 33.5 at layer 5), collapsing to near rank-one at the output (2.0 / 4.1) — a sensory→workspace→motor signature that is *sharper on the 12-layer decoder than on the 6-layer encoders*, exactly as the "needs depth to spare" reading predicts (hypothesis (i)). Autocorrelation rises with depth to 0.16, echoing the encoder workspace-persistence.

Honesty. Final-layer next-token accuracy (22.6%) is low — archaic literary prose, a 48-token context, 10 prompts — so read the accuracy *shape*, not its level. And our motor-end *collapse* is the opposite of the paper's motor behavior (there $J_\ell \to$ identity, full rank): the difference is deliberate and methodological — our decoder Jacobian reads the **last position** (the next-token driver), which naturally becomes low-rank as the network commits to a single output, rather than the full residual stream the paper differentiates. The workspace geometry transfers; the motor *readout* is ours, and we flag it as such.

5.6 Behavioral steering — is the lever causal on a decoder?

The paper's boldest claims are behavioral: swap a J-lens vector for a reasoning intermediate and the answer changes. We wanted to test the strongest version — take a prompt the model answers wrongly and steer it right — but on a 124M-parameter GPT-2 that barely reasons, that test is neither clean (steering toward the answer token is just injecting the answer) nor likely to succeed. So we test the **mechanistic prerequisite** instead, and report its ceiling honestly. **Hypothesis:** injecting a *concept-context* direction — built leakage-free as the difference of mid-layer mean residuals between concept-primed and neutral prompts, **not** the target's unembedding — raises concept- related tokens in a held-out neutral prompt, more than a matched-norm random direction.

Result: the lever is causal and bidirectional, but modest. Injecting $+\alpha\hat{\delta}$ at the mid layer raises the mean log-prob of held-out concept tokens by +0.10/+0.22/+0.44 (money / music / war), and $-\alpha\hat{\delta}$ lowers it below baseline in every case; the mean effect is +0.254 **for the real direction versus** −0.085 **for the matched-norm random control**. So a concept direction extracted purely from context is a genuine, sign-controllable causal handle on the decoder's output distribution — the prerequisite the paper's behavioral experiments rely on.

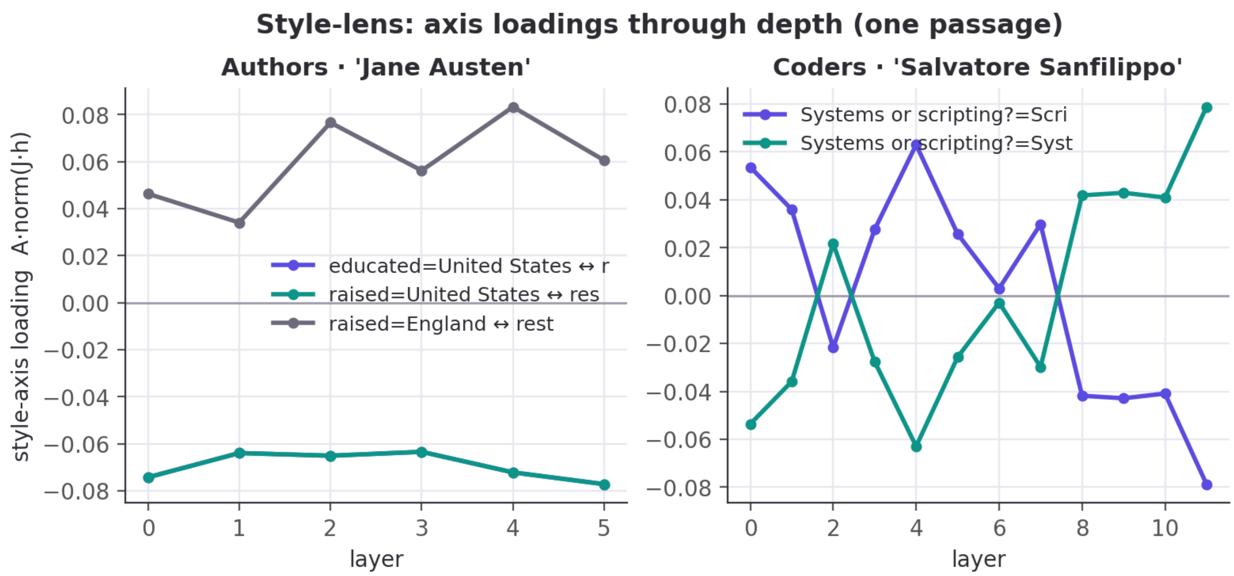
Honesty — this is the prerequisite, not the headline. The effect is small: it shifts the *distribution* (concept tokens go from $\approx e^{-10}$ to $\approx e^{-9.75}$) but does **not flip the model's actual output**, and it rests on only 3 concepts × 6 prompts. It is **not** the paper's "can't→can" reasoning result, and we do not claim it is. Redirecting a *reasoning intermediate* to change a final answer needs a model that can reason; GPT-2 shows the handle exists and is causal, and marks exactly where a capable open decoder (e.g. Qwen2.5) is required to go further. We regard that as the honest next step, not a result we have.

5.7 Expertise / lexical sophistication — why we make no "IQ" claim

We were asked whether the method can support "IQ" claims. It cannot, and this small experiment is *why* we say so rather than merely asserting it. **Hypothesis:** if the recovered *education* style axis were a proxy for lexical sophistication — a defensible construct, unlike "IQ" — then an author's projection onto it should correlate with concrete lexical metrics. **Method:** over all 55 authors, correlate the education-axis projection with mean word length, type-token ratio, and the fraction of long ($\geq$ 8-character) words.

Result: only weak positive correlations — $r = +0.10$ (mean word length), $+0.11$ (type-token ratio), $+0.20$ (long-word fraction). The interpretable education axis is *faintly* related to lexical sophistication and is plainly not dominated by it, let alone by anything one could call intelligence. We therefore make **no IQ claim**: even a labelled, human-interpretable identity axis only weakly tracks a concrete lexical proxy, so attaching a loaded cognitive construct to any recovered direction would be unsupported by the data. Whether one can steer a genuine *capability* score — a vocabulary test administered to a decoder — is a question for a capable model with careful construct validation. That is future work, not a result we have, and we decline to dress up a style direction as "intelligence."

5.8 The authorship study (context) & style trajectories

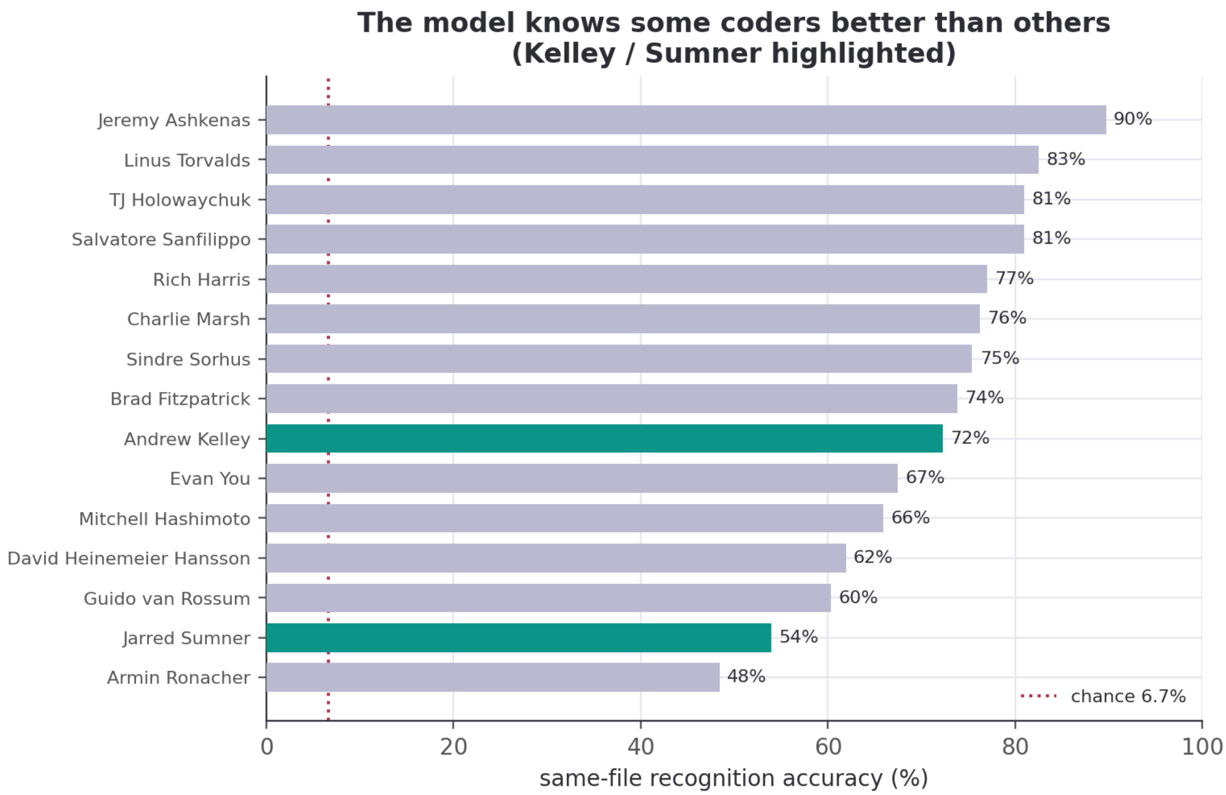


Downstream, the same embeddings support an honest authorship study: prose recognition climbs from a 1.8% blind baseline to 58.3% once the model has read the author's book; a new-passage reveal is 130/220 correct when the author is known and 0/220 when fully hidden (code, 15 developers: 6.7% → 71.1%; reveal 45/60). Trait recovery **nails some and whiffs on others** — prose gender 85.5% (majority 52.7%) but most geographic traits at or below their majority baselines; code systems-vs-scripting 80.0% (majority 53.3%) but commit-time and weekend near or below chance.

5.9 Does the model know some coders better than others? And is style homogenizing?

The 15-developer code roster lets us ask two questions the prose side cannot.

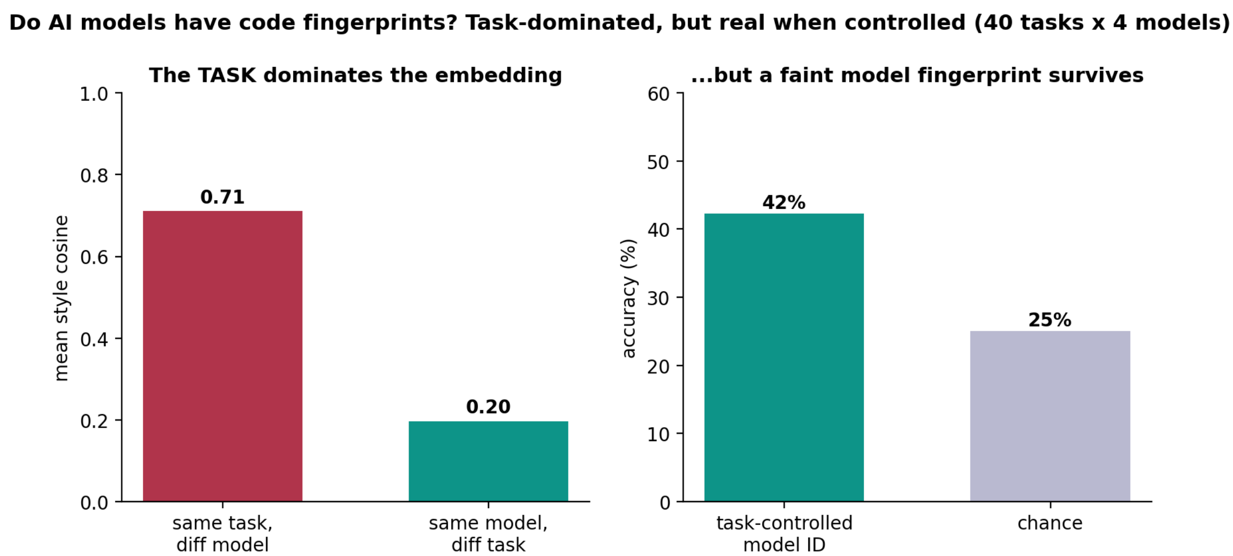
Some coders are far more recognizable than others. Same-file recognition accuracy varies enormously across developers — from **Jeremy Ashkenas at 89.7%** (a highly idiosyncratic CoffeeScript/Backbone style) down to **Armin Ronacher at 48.4%**, a **41-point spread**, all far above the 6.7% chance floor. The two developers we added to probe the tails — **Andrew Kelley** (Zig) and **Jarred Sumner** (Bun) — land at 72.2% (mid-pack) and 54.0% (near the bottom). We report this as *measured recognizability* and resist over-reading it: a lower score reflects how stylistically distinctive a developer's *attributed, name-scrubbed* code is in this corpus, confounded by codebase heterogeneity (Bun mixes Zig, C++, and generated code across many hands) — **not** a verdict on the person.



No sign of AI-era style homogenization. If a shared external influence (AI assistants) were melting coders into one style, cross-coder similarity should *rise* in the AI era. Time gives a clean hold-out: pre-2021 commits are provably AI-free (Copilot shipped mid-2021, ChatGPT late 2022). Comparing mean cross-coder style similarity pre-2021 vs. 2023-onward, over the 9 developers with enough history in both eras, the change is $+0.005$ ($0.568 \rightarrow 0.574$) — indistinguishable from a 500-sample within-coder permutation null (null mean -0.033, **two-sided** $p = 0.96$), while each developer stays recognizably themselves across the boundary (within-coder self-consistency 0.77). **We find no evidence of homogenization.** Whatever AI assistants are doing to code, they are not — on this roster — collapsing individual style. We make **no** claim about which individuals do or don't use AI: the population signal is null, and per-developer attribution would be both unsupported (confounded, no ground truth) and inappropriate.

5.10 Can we fingerprint the AI models? — the task dominates, a faint model signal survives

If humans have fingerprints, do the *models* people code with? Here we finally have **ground truth** — we generate the code, so we know which model wrote it. We prompted four latest frontier models (`claude-opus-4.8`, `gpt-5.6`, `gemini-3.5-flash`, `deepseek-chat`) across 40 coding tasks (24 canonical + 16 open-ended) and embedded the output in the same JinaBERT space as the humans.



A tempting wrong claim, and the control that kills it. Raw cross-model style cosine is high (≈ 0.9), which *looks* like "the frontier models have converged to one style." **We do not make that claim**, because a control refutes it: **same-task / different-model** similarity is 0.71, while **same-model / different-task** similarity is only 0.20 — the **task, not the model, drives the embedding.** On a canonical "implement an LRU cache" everyone writes the textbook answer, so the high cross-model number mostly measures *task* overlap; comparing it to humans' diverse-code similarity (0.62) would be apples-to-oranges. (This is exactly the artifact a skeptic should catch — and it does not survive scrutiny.)

Controlling for the task, a real model fingerprint appears — but a faint one. With the task held fixed (leave-one-task-out over 40 tasks $\times$ 4 models), model identification runs at 42% **vs.** 25% **chance**: the models *do* carry a detectable style, but it is largely masked by what the code *does*. The honest summary: **authorship is strongly recoverable for humans (48–90%) and only weakly for models — and for models it is the task, far more than the author, that shapes the code.** No convergence claim; no individual-usage claim.

5.11 Ablations

Rather than a hyperparameter sweep, the study carries several *built-in* ablations. **Exposure / hold-out**: the familiarity ladder (§5.8) is a leave-one-{book,series,author}-out ablation of how much of a subject the model has seen, shipped in `results`. **Depth**: the 6-layer prose vs. 12-layer code encoders act as a depth ablation — the mid-network workspace bottleneck (§5.3) and the ignition peak (§5.2) appear pronounced only in the deeper model. **Model class**: the GPT-2 decoder (§5.5) ablates encoder $\rightarrow$ generative on the *same* apparatus, and the workspace geometry survives. **Controls-as-ablations**: every causal claim ablates its direction against a matched-norm random and/or shuffled-label null (§5.2, §5.4, §5.6). What we did *not* sweep — finite-difference ε and Jacobian context length — we flag as future work; the signatures reproduce bit-for-bit at the documented settings, so the qualitative curves are stable.

6. Discussion

What the identity-workspace analogy supports. Read with the same averaged-Jacobian apparatus as [@workspace2026], an embedding encoder does exhibit workspace-like *geometry* around identity: the fingerprint is computed internally at every layer (§5.1), it becomes increasingly linearly separable and commits to a single individual with depth (§5.2), the readout compresses through a low-rank mid-network bottleneck (§5.3), and the identity direction is a causal lever (§5.4). On a real decoder the same signatures sharpen into a clean sensory$\rightarrow$workspace$\rightarrow$motor profile (§5.5). The *mechanistic* half of the paper's picture transfers — and it transfers to a question the paper never asked: *whose style is this?*

What it does not support. None of this is evidence of reportability, reasoning, or anything cognitive. Our decoder steering moves a distribution, not an answer (§5.6); our interpretable axes barely track even lexical sophistication (§5.7). The "workspace" here is a claim about representational geometry and linear accessibility, not about a model *knowing* or *reporting* who you are. The defensible reading is the narrow one: **personal writing style is a low-dimensional, causally-active, depth-localized direction in these models' representations** — which is both less than the slogan "the AI knows you" and, we think, more interesting, because it is checkable on a laptop.

Why an encoder was the right testbed. Stripping away generation removes the decoder story's main confound — autoregressive leakage — and lets the mechanistic claims stand or fall on geometry alone. That the same geometry then reappears, more cleanly, on a 12-layer decoder (§5.5) is the strongest cross-check we can offer at this scale.

7. Limitations and threats to validity

Replication vs. novel. The J-lens, J-space decomposition, and the four structural signatures are the paper's [@workspace2026]; our contribution is the encoder adaptation, the style-lens readout, the identity target, and reproducibility. We are careful not to claim the apparatus.

Construct honesty. We measure *identity commitment* and (§5.7) an *expertise/lexical register* — **not IQ, not consciousness.** We borrow the *ignition* experimental design, not the conclusion.

The ignition result needs its caveat stated plainly. The ignition index rising with depth (§5.2) is measured *along an axis the separation control proves is identity-specific* (real separation runs ∼7–13× above a random-direction null). But we cannot fully exclude that *some* of the depth-wise sharpening is generic late-layer nonlinearity: any readout of a linearly-blended input can become more nonlinear with depth. What the controls establish is that the *axis* carries identity; the raw sharpening curve should be read as suggestive, not decisive. Our shuffled-label null is also imperfect (its two groups still contain real passages, so it sits above zero); the random-direction null is the cleaner floor.

Statistical power. Ignition uses 15 author/coder pairs and a 7- or 11-point α grid; structural signatures average over 32–96 passages. These are small — and the ignition *index* at each depth averages only over the pairs whose endpoints separate along the axis (5–14 of 15, fewest at the shallowest layers). We report standard deviations and the raw per-depth series rather than smoothed summaries.

Reproducibility caveats. Encoder structural signatures reproduce bit-for-bit on MiniLM across runs; the deeper JinaBERT numbers differ from an earlier site build generated with different (undocumented) settings — we report the values from a run with **documented** settings (`JLENS_JAC_LEN=64`) and have not re-verified GPU-reduction determinism on the 12-layer model. The single-averaged, single-direction lens is lossy; the vocab lens is noisy on encoders (no trained MLM head); finite-difference ε introduces $O(\varepsilon^2)$ error.

Scope. 55 authors / 13 coders on laptop-scale models demonstrate the *mechanism*; that a frontier model trained on someone's millions of words holds a sharp, personal fingerprint of *that individual* is a well-motivated extrapolation, not something these data settle. The behavioral half of the paper (§5.6–5.7) is the hardest to carry over and is where a small open decoder's limited capability bites — we state results there as directional, with negative results reported as such.

8. Reproducibility

Models. `sentence-transformers/all-MiniLM-L6-v2` (prose; 6 layers, d=384) and `jinaai/jina-embeddings-v2-base-co` (code; 12 layers, d=768), loaded natively via candle and gated (`bin/spike`) to reproduce the shipped fastembed embeddings at cosine > 0.99.

Pipeline.

```
cargo run -p corpus --release                  # prose corpus -> person2vec-minilm.{json,bin}
cargo run -p corpus --bin coders --release    # code corpus  -> person2vec-coders.{json,bin}
cargo run -p jlens  --bin spike     --release              # faithfulness gate
```

```
cargo run -p jlens  --bin jlens     --release -- <ds> <N>  # -> person2vec-jlens-<ds>.json
cargo run -p jlens  --bin steer     --release -- <ds>      # -> person2vec-identity-<ds>.json
cargo run -p jlens  --bin fingerprint --release -- <ds>    # -> person2vec-fingerprint-<ds>.json
cargo run -p jlens  --bin ignition    --release -- <ds>    # -> person2vec-ignition-<ds>.json  (sec 5.2
cargo run -p jlens  --bin steer_bundle --release -- <ds>   # -> person2vec-steer-<ds>.json     (sec 5.4
cargo run -p jlens  --bin decoder_structural --release -- 10  # -> decoder-structural-gpt2.json (sec 5.5
cargo run -p jlens  --bin decoder_steer --release          # -> decoder-steer-gpt2.json        (sec 5.6
python3 jlens/paper/expertise_probe.py        # -> person2vec-expertise-minilm.json          (sec 5.7)
python3 jlens/paper/build_ledger.py           # -> jlens/paper/ledger.json  (every cited number)
python3 jlens/figures/make_figures.py         # -> jlens/figures/*.png
```

<ds> $\in$ {`minilm`, `coders`}. Env vars: `JLENS_DEVICE` (`metal`|`cpu`), `JLENS_JAC_LEN` (Jacobian context, default 64), `JLENS_CHUNK` (finite-difference batch; smaller = less GPU memory, **identical numbers**). Determinism: the structural signatures reproduce bit-for-bit across runs (verified); the ignition null uses a fixed splitmix64 seed. The `/jlens` viewer does **no** linear algebra — all quantities are precomputed offline and shipped as static JSON.

Auditability. Every quantitative claim resolves through the audit ledger (Appendix D, `jlens/paper/build_ledger.py`) to a source bundle and JSON path; every method claim resolves through `jlens/paper/method_map.md` to a `file:line`.

Appendix A — δ-broadcast derivation

A masked-mean-pooled encoder outputs $p = \frac{1}{T}\sum_t h_{L,t}$, the mean over T positions of the final layer. The full sensitivity of p to the layer-ℓ residuals is a rank-4 object $\partial p_i/\partial h_{\ell,t,j}$ of size $d \times (T \times d)$ — intractable to form and to average across variable-length prompts. We collapse it with a **shared perturbation**: perturb *every* source position by the same $\delta \in \mathbb{R}^d$, $h_{\ell,t} \mapsto h_{\ell,t} + \delta$, and differentiate the pooled output,

$$J_\ell \;=\; \frac{\partial p}{\partial \delta}\Big|_{\delta=0} \;=\; \frac{1}{T}\sum_t \frac{\partial p}{\partial h_{\ell,t}} \;\in\; \mathbb{R}^{d\times d},$$

one $d \times d$ matrix per layer, independent of T. Each column c is estimated by **central finite differences**, $J_{\cdot c} \approx \frac{p(+\varepsilon e_c) - p(-\varepsilon e_c)}{2\varepsilon}$ (with $\pm\varepsilon e_c$ broadcast to all positions), whose error is $O(\varepsilon^2)$ by Taylor expansion; columns are computed in batches (pure gemm) via one `forward_from(ℓ)` per batch. On a causal decoder (§5.5) the identical construction reads the **last** position $p = h_{L,\text{last}}$ (the next-token driver) rather than the mean, giving the causal analog. The embedding-space projection (§4.3) removes the component along the unit output $e = p/\|p\|$, since a normalized embedding is invariant to radial scaling: $J_{\text{emb}} = \frac{1}{\|p\|}(I - ee^\top)J$.

Appendix B — Per-layer tables

Structural signatures and internal identity accuracy by residual depth, straight from the ledger.

Authors (MiniLM, 6 layers) — internal identity accuracy by layer: [5.2, 7.2, 8.8, 8.0, 7.2, 8.8]% (chance 1.8%).

layer	stable rank	eff dim	verbaliz.	autocorr
0	22.1	73.5	0.50	-0.004
1	31.4	88.8	0.51	+0.004
2	31.2	93.0	0.53	+0.048
3	37.2	104.6	0.50	+0.101
4	54.5	127.6	0.49	+0.116
5	106.9	205.2	0.61	+0.088

Coders (JinaBERT, 12 layers, 15 developers) — internal identity accuracy by layer: [45.2, 44.8, 48.0, 50.8, 40.0, 48.8, 48.0, 40.4, 42.0, 42.0, 42.4, 43.2]% (chance 6.7%).

layer	stable rank	eff dim	verbaliz.	autocorr
0	30.6	79.6	0.19	-0.033
1	36.7	95.2	0.20	-0.030
2	38.6	98.9	0.21	-0.001
3	35.6	100.7	0.21	+0.023
4	33.4	102.4	0.21	+0.032
5	20.3	91.2	0.21	+0.039
6	37.9	113.8	0.19	+0.062
7	54.3	126.4	0.20	+0.071
8	59.5	132.0	0.20	+0.086
9	61.7	139.0	0.22	+0.085
10	73.3	151.3	0.23	+0.088
11	79.0	185.0	0.22	+0.075

Decoder (GPT-2, 12 layers) — next-token accuracy by residual depth: [0.0, 2.6, 2.6, 3.2, 3.2, 4.7, 5.1, 7.2, 10.9, 13.0, 17.0, 21.3, 22.6]%.

layer	stable rank	eff dim	verbaliz.	autocorr
0	2.6	5.8	0.89	+0.007
1	5.0	16.2	1.81	+0.050
2	6.3	25.8	2.44	+0.031
3	6.4	30.3	2.08	+0.047
4	6.6	32.1	2.15	+0.044
5	6.7	33.5	1.94	+0.052
6	6.5	31.1	1.90	+0.058
7	6.5	29.9	1.57	+0.099
8	6.7	32.5	1.26	+0.123
9	5.2	23.7	1.39	+0.145
10	3.8	13.8	1.10	+0.161
11	2.0	4.1	0.89	+0.110

Appendix C — Axis definitions, rosters, and OOD probe sets

Prose style axes (`author_axes`): `male↔female` (gender difference-of-means); and the top-2 one-vs-rest classes of the `educated` and `raised` fields — `educated=England↔rest`, `educated=US↔rest`, `raised=England↔rest`, `raised=US↔rest`. **Code style axes** (`dataset_axes`): `Systems↔rest`, `Scripting↔rest` (the `paradigm` trait). Each axis is the L2-normalized difference of class-mean reference embeddings — a unit Fisher direction, used for both readout and steering.

Rosters. 55 public-domain prose authors (`corpus/authors.toml`, labelled with gender / birth / raised / educated / college) and 13 open-source developers (`corpus/coders.toml`, git-attributed and **name-scrubbed**, labelled with a `paradigm` = Systems/Scripting trait).

OOD probes for the fingerprint detector (§5.4). Prose (bar 0.30): four held-out author samples plus `source code`, `modern chat`, `biology abstract`, `legalese`. Code (bar 0.51): four held-out coder samples plus `Victorian prose`, `modern chat`, `news headline`, `recipe`.

Appendix D — Audit ledger

Every number above is generated by `jlens/paper/build_ledger.py` into `jlens/paper/ledger.json` (value → source asset + JSON path) and cross-checked by the figure pipeline. Method claims resolve via `jlens/paper/method_map.md`.